Owner's Manual: Northfen PW-3400G Gas Pressure Washer

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Important safety information

Read and understand every safety message before you set up or operate the Northfen PW-3400G Gas Pressure Washer.

Signal words

Safety messages in this manual use these signal words:

DANGER	A hazardous situation that, if not avoided, will cause death or serious injury.
WARNING	A hazardous situation that, if not avoided, could cause death or serious injury.
NOTICE	A situation that could damage the machine or other property but will not cause injury.

Safety messages for this machine

⚠ DANGER

Carbon monoxide.

Engine exhaust contains carbon monoxide, a poisonous gas you cannot see or smell. Breathing it can kill you in minutes.

Run the engine outdoors only, at least 15 ft (4.6 m) from windows, doors and vents.

Never run the engine in a garage, basement, shed or other enclosed space, even with the door open.

⚠ WARNING

Injection hazard from high-pressure spray.

The spray can cut skin and inject water and chemicals into the body. An injection can look like a small cut but can cause serious injury or loss of a limb.

Never point the spray gun at yourself, other people or animals.

Never put your hand or fingers in front of the nozzle.

Engage the trigger lock whenever you are not spraying.

If spray pierces the skin, get emergency medical care right away. Tell the doctor it is an injection injury.

⚠ WARNING

Fire and explosion.

Gasoline and its vapors are highly flammable. A spark or hot surface can ignite spilled fuel.

Stop the engine and let it cool for at least 2 minutes before you add fuel.

Fill the tank outdoors, away from flames and sparks.

Wipe up spilled fuel before you start the engine.

⚠ WARNING

Stored pressure.

The hose and spray gun stay pressurized after you stop the machine. Disconnecting a pressurized part can release a burst of water and cause injury.

Relieve pressure before you disconnect the hose, change a nozzle, or service the machine.

Personal protective equipment

Wear this equipment every time you operate the machine:

- Safety glasses or goggles that meet ANSI Z87.1
- Closed-toe, slip-resistant shoes
- Long pants
- Hearing protection

How the pressure washer works

The PW-3400G takes water from a standard garden hose and uses a pump to raise its pressure to as much as 3,400 psi (234 bar). The spray gun and nozzle turn that pressure into a focused stream that removes dirt, mildew and loose paint.

The path water takes

Water enters the pump through the garden hose inlet. A screen in the inlet catches debris that could damage the pump. The gasoline engine drives the pump, which pushes the water through the high-pressure hose to the spray gun. The nozzle on the end of the spray wand narrows the stream and sets the spray pattern.

What happens when you release the trigger

When you release the trigger, water has nowhere to go. An *unloader valve* senses the rise in pressure and sends the water back through the pump in a loop. This is called *bypass mode*. On the gas model, the engine keeps running in bypass mode.

Water that loops in bypass mode heats up. A *thermal relief valve* releases hot water from the pump before it can damage the seals. If you see water dripping from under the pump after a long pause, the valve is working as designed.

Main components

Garden hose inlet	Connects the water supply. Contains the inlet screen.
Pump	Triplex plunger pump with serviceable oil. Built for daily commercial use.
Engine	212 cc overhead-valve engine with recoil start and low-oil shutdown.
High-pressure hose	Carries pressurized water from the pump to the spray gun.
Spray gun and wand	The trigger starts and stops the spray. The trigger lock prevents accidental spraying.
Quick-connect nozzles	Color-coded tips that set the spray angle.

Choosing a nozzle

A narrower spray angle delivers more force to a smaller area. Start with the widest nozzle that does the job, and move to a narrower one only if needed.

Color	Angle	Typical use
Red	0°	Small, stubborn spots on concrete or metal. Never use on wood, siding or vehicles.
Yellow	15°	Stripping loose paint and cleaning concrete.
Green	25°	General cleaning of decks, driveways and patio furniture.
White	40°	Rinsing vehicles, windows and siding.
Black	65°	Applying detergent at low pressure.

Set up and start the pressure washer

Connect the water supply and hoses, purge air from the system, and start the PW-3400G.

You need:

- A garden hose no longer than 50 ft (15 m) and a water supply of at least 3.0 gal/min (11.4 L/min)
- Fresh unleaded gasoline, 87 octane or higher, with no more than 10% ethanol
- The personal protective equipment listed in Important safety information on page 3

⚠ DANGER

Carbon monoxide.

Engine exhaust contains carbon monoxide, a poisonous gas you cannot see or smell. Breathing it can kill you in minutes.

Run the engine outdoors only, at least 15 ft (4.6 m) from windows, doors and vents.

Never run the engine in a garage, basement, shed or other enclosed space, even with the door open.

Place the machine on a level surface outdoors, where spray and runoff cannot reach electrical outlets or open drains.

1. Check that the inlet screen is clean and seated in the garden hose inlet.

   Rinse the screen under running water if you see debris.
2. Check the engine oil level.

   Remove the dipstick, wipe it, and reinsert it without screwing it in. The oil should reach the upper mark. The engine ships with oil, but the level can drop in transit.
3. **⚠ WARNING**

Fire and explosion.

Gasoline and its vapors are highly flammable. A spark or hot surface can ignite spilled fuel.

Stop the engine and let it cool for at least 2 minutes before you add fuel.

Fill the tank outdoors, away from flames and sparks.

Wipe up spilled fuel before you start the engine.

Fill the fuel tank to the bottom of the filler neck.

4. Connect the garden hose to the garden hose inlet and tighten it by hand.
5. Connect the high-pressure hose to the pump outlet and to the spray gun.

   Pull back on each quick-connect fitting to confirm it is locked.
6. Insert a nozzle into the end of the spray wand.

   For a first test, use the white 40° nozzle. See Choosing a nozzle on page 5.
7. **NOTICE**

   Pump damage.

   Running the pump without water overheats the seals in seconds.

   Turn on the water supply and purge air from the system before you start the machine.

   Turn on the water supply fully. Release the trigger lock, point the spray gun at the ground, and squeeze the trigger until a steady stream of water flows with no air sputtering.

   Air is purged from the pump and hoses.
8. Prepare the engine to start.
   a) Turn the fuel valve to **ON**.
   b) If the engine is cold, move the choke lever to **CHOKE**.
   c) Move the engine switch to **ON**.
9. Squeeze the trigger and hold it, then pull the starter cord slowly until you feel resistance, and pull it quickly.

   Holding the trigger releases pressure in the pump so the engine is easier to start.
10. After the engine warms up, slowly move the choke lever to **RUN**.
11. **⚠ WARNING**

    Injection hazard from high-pressure spray.

    The spray can cut skin and inject water and chemicals into the body. An injection can look like a small cut but can cause serious injury or loss of a limb.

    Never point the spray gun at yourself, other people or animals.

    Never put your hand or fingers in front of the nozzle.

    Engage the trigger lock whenever you are not spraying.

    If spray pierces the skin, get emergency medical care right away. Tell the doctor it is an injection injury.

    Hold the spray gun with both hands and squeeze the trigger to begin cleaning.

    Start with the nozzle at least 24 in. (60 cm) from the surface and move closer slowly. Engage the trigger lock whenever you pause.

The pressure washer is running and ready for use. When you finish, follow Shut down, relieve pressure and store the pressure washer on page 7.

Shut down, relieve pressure and store the pressure washer

Stop the PW-3400G, release trapped pressure, and drain the pump so the machine is safe to handle and ready for storage.

Follow these steps every time you finish using the machine, and before you change a nozzle, clean the inlet screen, or perform any maintenance.

Note: If you plan to store the machine for more than 30 days, add fuel stabilizer at your last refill of the season, then run the machine for at least 5 minutes so the treated fuel reaches the carburetor.

⚠ WARNING

Stored pressure.

The hose and spray gun stay pressurized after you stop the machine. Disconnecting a pressurized part can release a burst of water and cause injury.

Relieve pressure before you disconnect the hose, change a nozzle, or service the machine.

1. Engage the trigger lock on the spray gun.
2. Move the engine switch to **OFF** and turn the fuel valve to **OFF**.
3. Turn off the water supply.
4. **⚠ WARNING**

   Injection hazard from high-pressure spray.

   The spray can cut skin and inject water and chemicals into the body. An injection can look like a small cut but can cause serious injury or loss of a limb.

   Never point the spray gun at yourself, other people or animals.

   Never put your hand or fingers in front of the nozzle.

   Engage the trigger lock whenever you are not spraying.

   If spray pierces the skin, get emergency medical care right away. Tell the doctor it is an injection injury.

   Point the spray gun at the ground, release the trigger lock, and squeeze the trigger until water stops flowing.

   Pressure is relieved from the hose and spray gun.
5. Engage the trigger lock again.
6. Disconnect the garden hose, the high-pressure hose, and the spray gun.
7. Drain the water from the pump, hoses and spray gun.

   Tip the machine so the pump outlet faces down, and hold each hose vertically until it stops dripping.
8. ***NOTICE***

   Freeze damage.

   Water left in the pump can freeze, crack the pump and void the warranty.

   Drain all water and add pump protector before you store the machine where temperatures can fall below 32 °F (0 °C).

If you will store the machine for more than 30 days or below freezing, add pump protector.

Attach the pump protector bottle to the garden hose inlet and squeeze until foam comes out of the pump outlet.

9. Store the machine and accessories indoors in a dry place.

The machine is depressurized, drained and protected for storage.

Specifications

Technical data for the Northfen PW-3400G Gas Pressure Washer.

Table 1: Specifications for the PW-3400G

Specification	Value
Maximum pressure	3,400 psi (234 bar)
Flow rate	2.6 gal/min (9.8 L/min)
Engine	212 cc OHV, recoil start, low-oil shutdown
Fuel tank capacity	0.95 gal (3.6 L)
Engine oil	SAE 10W-30, 20 oz (0.6 L)
Pump	Triplex plunger, 7 oz (0.2 L) pump oil
High-pressure hose	50 ft (15.2 m), steel-braided
Nozzles included	0°, 15°, 25°, 40°, 65° (soap)
Maximum inlet water temperature	104 °F (40 °C)
Weight	74 lb (33.6 kg)

Troubleshooting

Find the symptom in the table, then try each solution in order. If the problem continues, contact an authorized service center.

⚠ WARNING

Stored pressure.

The hose and spray gun stay pressurized after you stop the machine. Disconnecting a pressurized part can release a burst of water and cause injury.

Relieve pressure before you disconnect the hose, change a nozzle, or service the machine.

Before you inspect or service any part, complete Shut down, relieve pressure and store the pressure washer on page 7 through the step that relieves pressure.

Table 2: Common problems and solutions

Problem	Possible cause	Solution
Engine does not start	Fuel valve or engine switch is off	Turn the fuel valve and engine switch to **ON**.
	Oil level is low	Add oil to the upper mark on the dipstick. The low-oil shutdown prevents starting until the level is correct.
	Pressure is trapped in the pump	Squeeze the spray gun trigger while you pull the starter cord.
Low or pulsing pressure	Not enough water supply	Open the water supply fully. Use a hose no longer than 50 ft (15 m) and check it for kinks.
	Air in the system	Stop the machine and purge air as described in Set up and start the pressure washer on page 5.
	Nozzle is worn or partly blocked	Clear the nozzle with the cleaning pin, or replace it.
Water leaks at a hose connection	Connection is loose or O-ring is damaged	Tighten the connection by hand. Replace the O-ring if the leak continues.
Water drips from under the pump	Thermal relief valve is releasing hot water	This is normal after a long pause in bypass mode. Squeeze the trigger more often, or stop the machine during long breaks.
Detergent does not flow	High-pressure nozzle is installed	Install the black 65° soap nozzle. Detergent flows only at low pressure.